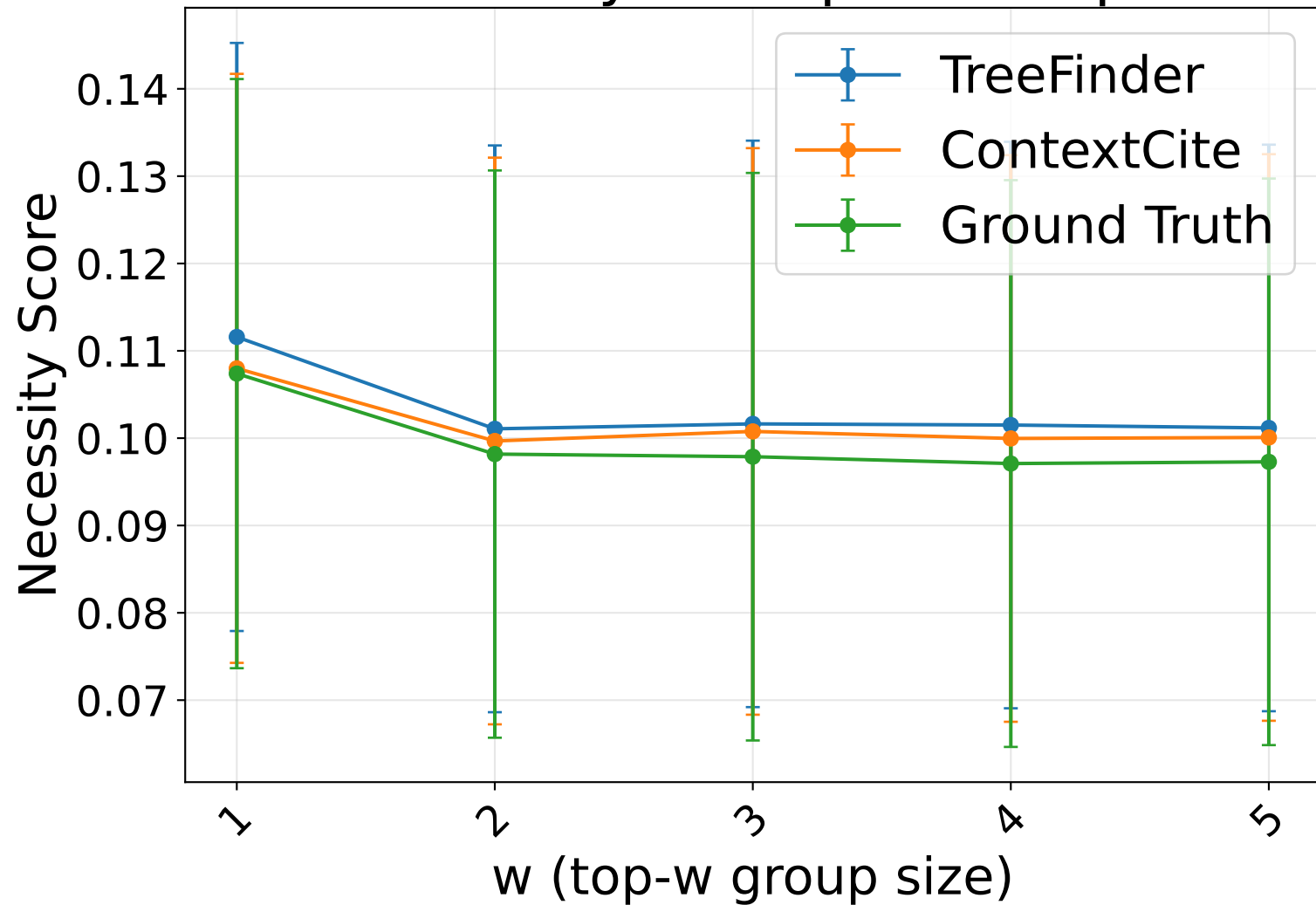


# Necessity for Top-w Groups



# Sufficiency for Top-w Groups

